🎭 AI Theater Planner

Personalized Theater Recommendation & Planning Agent

MATH 690 - Graduate Seminar : Agentic AI Project
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📋 Agenda

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🎯 Problem Statement

Current Challenges

Theater enthusiasts face multiple pain points:

- 📅 Hard to track which plays are showing when
- 🗺️ Difficult to filter by location/distance
- ⭐ No personalized recommendations based on past preferences
- 🎥 Missing context (actor interviews, reviews)
- ⏰ Manual calendar coordination

Our Solution

An AI agent that:

- Syncs with your calendar
- Recommends plays based on your taste
- Filters by location and availability
- Suggests pre/post-show content (interviews, reviews)

💡 Solution Overview

Core Features

1. Calendar Integration

- Automatically finds free time slots
- Avoids scheduling conflicts

2. Intelligent Filtering

- Location-based (max 30 min travel)
- Genre preferences (drama vs. comedy)
- Venue preferences (state theater vs. private)

3. Personalized Scoring

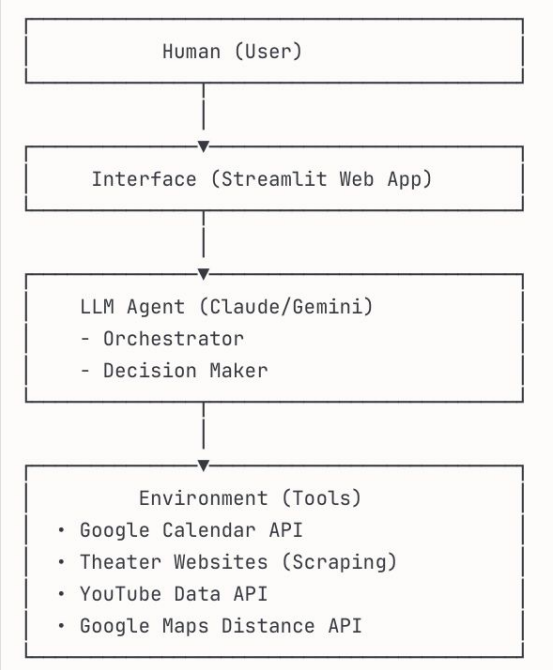
- Analyzes past ratings
- Predicts enjoyment likelihood

4. Content Discovery

- Actor interviews (YouTube)
- Reviews and articles
- Behind-the-scenes content

🏗️ Technical Architecture

High-Level Design (From GSU - Agents 2 Lecture Slide 16)



🎨 Prompting Techniques Used

1. Zero-Shot (from Slide 6)

Use Case: Simple queries

``` Examples: "What plays are showing today in Istanbul?" ```

2. Few-Shot (from Slide 7)

Use Case: Preference prediction

``` Examples:

- "Hamlet" → Liked (Drama, Classic)

- "Comedy Night" → Disliked (Shallow humor)

Predict: Will user like "King Lear"?

```

3. Chain-of-Thought (from Slide 8)

Use Case: Multi-step reasoning

``` 1. Analyze user's calendar 2. Find available time slots 3. Filter plays by location 4. Score based on past preferences 5. Rank and recommend```

🔧 System Components

Tool Calling (Slide 12)

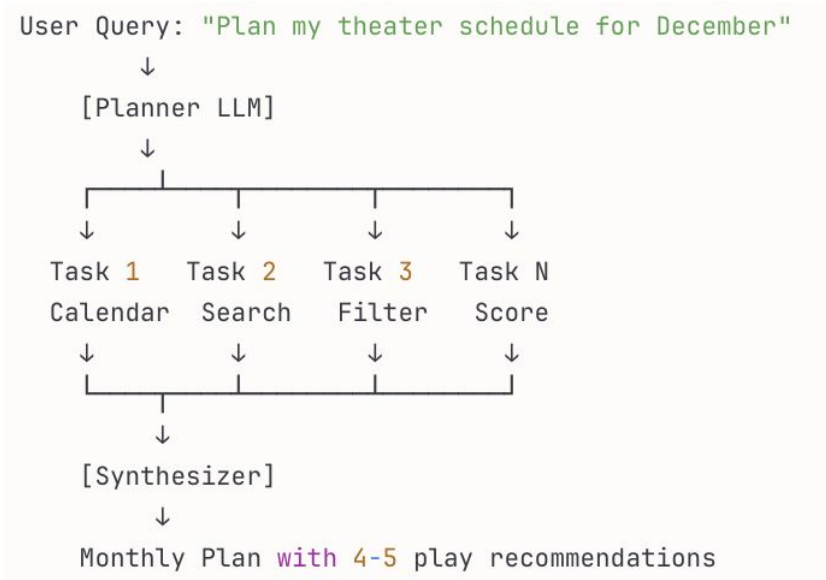
Possible Available Tools:

```
# Calendar Management
get_calendar_events(date_range)

# Theater Data
search_plays(city, date)
get_play_details(play_id)

# Location Services
calculate_distance(user_location, venue)

# Content Discovery
search_interviews(play_name, actor)
get_reviews(play_name)

# User Preferences
get_user_ratings(user_id)
save_user_rating(play_id, rating)
```

🔧 System Components

🧠 Agentic Architecture Pattern

Planner Pattern (Slide 23)



**Why Planner?
- Complex task requiring multiple steps
- Tasks can be executed in sequence
- Plan needs to be tracked across iterations

🔧 System Components

🔄 Reflection Loop (Slide 21)

Continuous Improvement

```
User: "Recommend a play"
    ↓
[Generator]: "I suggest 'Hamlet'"
    ↓
[Reflector]: "Check if user likes tragedy"
    ↓
User Feedback: ⭐⭐⭐⭐⭐
    ↓
[Update Model]: Increase weight for tragedy genre
```

**Benefits:
- Learns from user feedback
- Improves over time
- Self-correcting system

🔧 System Components

🛠️ Technical Stack

Backend

- **Language:** Python 3.11+

- **LLM Integration:** LiteLLM (multi-provider)

- **Models:** Claude 3.5 Haiku,Gemini 2.0 Flash

- **APIs:** - Google Calendar API - YouTube Data API v3 - Google Maps Distance Matrix API

🔧 System Components

🛠️ Technical Stack

Frontend

- **Framework:** Streamlit

- **Deployment:** Streamlit Cloud

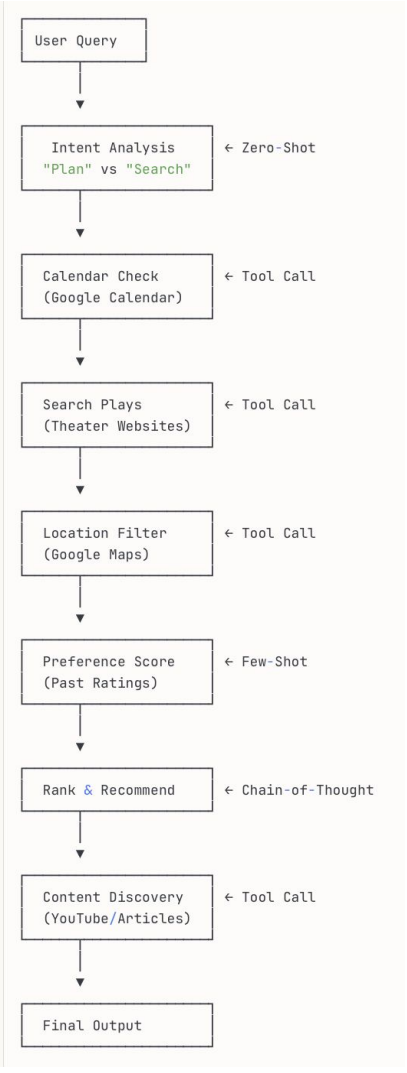
Data Storage

- **User Preferences:** SQLite

- **Cache:** Redis (for API responses)

🔧 System Components

📊 System Flow Diagram



Project &Risk Management

🚀 Future Work

Phase 1 (Current Project)

- ✅ Single city (Istanbul)
- ✅ Manual calendar sync
- ✅ Basic preferences

Phase 2

- 🔄 Multi-city support
- 🔄 International festivals
- 🔄 Travel cost estimation
- 🔄 Hotel/flight booking

Phase 3 (Advanced Features)

- 🎯 Group recommendations (friends)
- 🎟️ Automatic ticket purchasing
- 📱 Mobile app (React Native)
- 🌍 Multi-language support

🔢 Evaluation Metrics

Accuracy

- **Recommendation Hit Rate:** (% of recommended plays actually attended)

-**Target:** (i.e. 70% and above)

User Satisfaction

- **Average Rating:** (User feedback on recommendations)

- **Target:** (i.e. 4.2/5.0)

System Performance

- **Response Time:** (Time to generate plan)

- **Target:** (i.e. < 10 seconds)

Coverage

- **Play Database:** Number of plays tracked

- **Target:** (i.e. 50 plus active plays)

(Expected) Learning Outcomes

1. Agentic AI Patterns
    - Planner architecture
    - Tool calling integration
    - Reflection loops
2. Prompting Techniques
    - Zero-shot for quick queries
    - Few-shot for preference learning
    - Chain-of-thought for complex reasoning
3. Real-World Integration
    - API integration (Google Calendar, Maps)
    - Web scraping (theater sites)
    - Data persistence (user preferences)

References

Course Materials:

- Lecture Slides: "Prompting LLMs" (Week 3)
- Lecture Slides: "Architecture"(Week 4)

Technical Resources:

- Anthropic Claude Documentation
- LiteLLM Documentation - Google Calendar API v3
- YouTube Data API v3

Inspiration:

- Erdem Özcan

🙏 Thank You!

Contact

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🔗 Demo

Will be Available At [theater-planner-demo.streamlit.app]

💬 Q&A

Possible Common Questions:

Q1: How do you handle multiple users?

A1: Each user has isolated preference profile in database

Q2: What if play info is incorrect?

A2: Reflection loop allows users to report errors, system self-corrects

Q3: Privacy concerns with calendar access?

A3: OAuth2, read-only access, user can revoke anytime

Q4: Cost of API calls?

A4: Cached responses, batched requests, ~$5/month for 100 users